

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM} + S_1 + S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q}Y_u\tilde{H}u + \text{h.c.}$
Y_d	$-\bar{Q}Y_dHd + \text{h.c.}$
Y_e	$-\bar{L}Y_eHe + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F}M_FF$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 HHS_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$\beta_{g_1}^{(1)}$

Quantity: g_1

$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$

Column Representative term structure	
T1	$\frac{17g_1^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{17g_1^3}{2}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{17g_2^3}{6}$

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$-3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$
T2	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$
Quantity: Y_e

$16\pi^2\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-3\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_e)_{ij}$
T2	$+3\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_e)_{ij}$
T3	$+\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$3\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_e)_{ij}$	$+3\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_e)_{ij}$	$+\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_1^2}{2}$
T2	$-\frac{3g_1^2m_1^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	$+4\lambda_{12}^{(1)}m_2^2$

Quantity: m_1^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_1}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{S_1}^{(1)}m_1^2$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_2^2}{2}$
T2	$-\frac{27g_1^2m_2^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{27g_1^2m_2^2}{2}$	$+4\lambda_{12}^{(1)}m_1^2$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+6\lambda_{S_2}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{S_2}^{(1)}m_2^2$

$\beta_{\lambda_1}^{(1)}$

Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$
T3	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$
T2	$+4\lambda_5\,(\lambda_5)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$	$+4\lambda_5\,(\lambda_5)^*$	$+ \frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Quantity: λ_1

Terms 4 of 5

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+4\lambda_{HS_1}^{(1)2}$
T3	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)^2}$
T2	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_2}^{(1)^2}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$

$$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{9g_2^4}{4}$
T2	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3	$-9g_2^2\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+ \frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$
T3	$+12\lambda_{S_1}^{(1) 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_1}^{(1) \ 2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+12\lambda_{S_1}^{(1) 2}$

$\beta_{\lambda_{S_2}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{9g_2^4}{4}$
T2	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3	$-9g_2^2\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$	$-9g_2^2\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{243g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{243g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+\frac{\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$
T3	$+12\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$+12\lambda_{S_2}^{(1) \ 2}$

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$
T2	$+6\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_1}^{(1)}$
T2	$+\frac{3g_1^4}{4}$
T3	$-3g_1^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{HS_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_1}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+6\lambda_5\ (\lambda_5)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+6\lambda_5\ (\lambda_5)^*$	$+\frac{9g_2^4}{4}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_2}^{(1)}$
T2	$+\frac{27g_1^4}{4}$
T3	$-15g_1^2\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{27g_1^4}{4}$	$-15g_1^2\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4\lambda_{HS_2}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$

$\beta_{\lambda_{12}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$3\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$3\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{12}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{27g_1^4}{4}$
T2	$-15g_1^2\lambda_{12}^{(1)}$
T3	$+4\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{27g_1^4}{4}$	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+\frac{3\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_5}^{(1)}$
Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$
T2	$-9g_2^2\lambda_5$
T3	$-9g_1^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	$-9g_2^2\lambda_5$	$-9g_1^2\lambda_5$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$16\pi^2\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$
T3	$-9g_2^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Quantity: $\lambda^{(A)}_{HS_1}$

$$16\pi^2\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu}=\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_1}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T2	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$
T3	$+g_1^2\left(6g_2^2-3\lambda^{(A)}_{HS_1}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda^{(A)}_{HS_1}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(6g_2^2-3\lambda^{(A)}_{HS_1}\right)$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_1}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)} (6\, \text{Tr}\Big(Y_d\, (Y_d)^\dagger\Big) + 2\, \text{Tr}\Big(Y_e\, (Y_e)^\dagger\Big) + 6\, \text{Tr}\Big(Y_u\, (Y_u)^\dagger\Big))$
T2	$+16\lambda_5\, (\lambda_5)^*$
T3	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{HS_2}^{(A)} (6\, \text{Tr}\Big(Y_d\, (Y_d)^\dagger\Big) + 2\, \text{Tr}\Big(Y_e\, (Y_e)^\dagger\Big) + 6\, \text{Tr}\Big(Y_u\, (Y_u)^\dagger\Big))$	$+16\lambda_5\, (\lambda_5)^*$	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_{HS_2}^{(A)}$
T2	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

Quantity: 1, 1, 3, 3

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1134}}^{(1)}$

Quantity: 1, 1, 3, 4

$$16\pi^2\,_{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1144}}^{(1)}$

Quantity: 1, 1, 4, 4

$$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{2lambda T_3 \overline{y_1 y_2}}{3MF}$

$$16\pi^2\,\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$

Quantity: 1, 1, 5, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambda12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda12Inv2y_1y_2}{3MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$

Quantity: 1, 1, 5, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{i\lambda\,nv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{i\lambda\,nv2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$

Quantity: 1, 1, 6, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$

Quantity: 1, 1, 6, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$\beta_{C_{1213}}^{(1)}$

Quantity: 1, 2, 1, 3

$$16\pi^2\,\mu\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1214}}^{(1)}$

Quantity: 1, 2, 1, 4

$$16\pi^2\,_{\mu}\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1223}}^{(1)}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,_{\mu}\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta_{C_{1224}}^{(1)}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,\mu\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\lambda T_3 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\lambda T_3 \overline{y_1 y_2}}{3 M F}$

$$16\pi^2\,\mu\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\textit{lambda}12Inv2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 5, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{24MF}$

$$16\pi^2\,\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{i\lambda v^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{i\lambda v^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 5, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{24 M F}$

$$16\pi^2\,\mu\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 6, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{24 M F}$

$$16\pi^2\,\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 6, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{24MF}$

$$16\pi^2\,\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 7, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{24MF}$

$$16\pi^2\,\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 7, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{24 M F}$

$$16\pi^2\,\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{i\lambda v^2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_1^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{i\lambda v^2 y_1 y_2}{6 M F}$	$-\frac{\sqrt{3} i g_1^2 y_1 y_2}{4 M F}$	$-\frac{\sqrt{3} i g_2^2 y_1 y_2}{8 M F}$

Quantity: 1, 2, 8, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{24 M F}$

$$16\pi^2\,\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 8, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{24MF}$

$\beta_{C_{2211}}^{(1)}$

Quantity: 2, 2, 1, 1

$$16\pi^2\,_{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$$

Column	Representative term structure
T1	$\frac{lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2lambda T_3 \overline{y_1 y_2}}{3MF}$

$$\beta_{C_{2212}}^{(1)}$$

Quantity: 2, 2, 1, 2

$$16\pi^2\,_{\mu}\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{2222}}^{(1)}$

Quantity: 2, 2, 2, 2

$$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{2\lambda T_3 \overline{y_1 y_2}}{3MF}$

$$16\pi^2\,\mu\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda}12\textit{Inv}2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\textit{lambda}12\textit{Inv}2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$

Quantity: 2, 2, 7, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{i\lambda v^2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3}ig_1^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3}ig_2^2 y_1 y_2}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{i\lambda v^2 y_1 y_2}{3M F}$	$-\frac{\sqrt{3}ig_1^2 y_1 y_2}{2M F}$	$-\frac{\sqrt{3}ig_2^2 y_1 y_2}{4M F}$

Quantity: 2, 2, 7, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,_{\mu}\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambda12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{ilambda12Inv2y_1y_2}{3MF}$	$\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$

Quantity: 2, 2, 8, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$

Quantity: 2, 2, 8, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$